Vikalp Pandey

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EDUCATION

BHARATI VIDYAPEETH COLLEGE OF ENGINEERING — *Paschim Vihar, Delhi, India*
Bachelor of Technology in Electronics and Communication Engineering — *July 2025*

SCSD SARVODAYA VIDYALAYA — *Rohini Sector 9, Delhi, India*
Class 12 — *July 2021*

SD SARASWATI BAL MANDIR — *Rajiv Nagar, Delhi, India*
Class 10 — *April 2019*

WORK & LEADERSHIP EXPERIENCE

Company: SequelString AI — Mumbai, Delhi/India
Nov,2025 - Current

Project: *Metro Reconciliation Process*

- Built an end-to-end web application using **React, FastAPI, Python, and OpenPyXL** that helped the accounts team move repeated Excel work into one simple workflow.
- Automated the matching of **GSTR-2A invoices** with account transaction records by cleaning invoice numbers, combining tax values, and generating separate Excel sheets for matched and unmatched entries.
- Developed a **Metro bank statement upload flow** that reads XLSX files, validates transaction dates, narration, debit and credit values, duplicate rows, and running balances before calculating the closing balance.
- Designed **REST APIs** to create reconciliation runs, upload one or multiple statements, return file-level validation results, and track each run through clear **DRAFT, CALCULATED, and REJECTED** statuses.
- Added controls such as unique batch IDs, **SHA-256 checksums, duplicate-file checks, file-size and type validation, and read-only source storage** to keep original statements safe and traceable.
- Built **responsive React screens** for run creation, statement upload, status tracking, and file-level calculation results, with clear validation messages so users can quickly understand and correct problems.

ACADEMIC AND PERSONAL PROJECTS

PROJECT-1: StoreX - Backend-Driven Cloud Storage Platform

Tech Stack: *TypeScript, Node.js, Hono, MongoDB, Redis, AWS S3/SQS/Lambda, OpenFGA, Stripe*
Demo / GitHub / Link

- **Designed REST APIs and MongoDB models** for nested folders, paginated listings, search, recent items, trash/restore workflows, and storage-quota tracking.
- **Built a three-layer caching strategy** with TanStack React Query in the browser, CDN for distribution and Redis in the API, reducing duplicate REST calls and MongoDB reads while cache invalidation keeps results fresh.
- **Queued signup emails with Amazon SQS** so the API responds quickly while a background worker retries temporary delivery failures, keeping signup latency stable during traffic spikes.
- **Managed recurring storage subscriptions with Stripe** by recording each buyer's plan, amount, currency, and current status while verified webhooks synchronize billing-cycle events and storage quotas.
- **Handled files larger than 5 GB with multipart uploads** by splitting files into parts, tracking progress, and completing or aborting uploads reliably.
- **Implemented relationship-based authorization** with OpenFGA for file and folder sharing, inherited permissions, and Shared with me access.
- **Secured authentication flows** with JWT sessions, bcrypt hashing, email OTP verification, password resets, and Google/GitHub OAuth support.
- **Preparing AWS Lambda deployment with SST and Amazon CloudFront** so edge caching can serve content closer to users, reduce S3/API origin traffic, absorb traffic spikes, and lower global response latency.

SKILLS

Programming: TypeScript, JavaScript
Backend & Data: Node.js, Hono, MongoDB, PostgreSQL, Redis, REST APIs
Cloud & DevOps: AWS
Frontend: React, Axios, TanStack Query, Tailwind CSS
Developer Tools: Git, GitHub, Turborepo, SST